

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Each base of a right rectangular prism has a length of **19** inches and a width of **8** inches. The prism has a volume of **2,736** cubic inches. What is the height, in inches, of the prism?

Answer

- A. **18**
- B. **27**
- C. **144**
- D. **152**

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length of the base, w is the width of the base, and h is the height of the prism. It's given that each base of a right rectangular prism has a length of **19** inches and a width of **8** inches, and that the prism has a volume of **2,736** cubic inches. Substituting **19** for ℓ , **8** for w , and **2,736** for V in the formula $V = \ell wh$ gives **2,736** $= (19)(8)(h)$, or **2,736** $= 152h$. Dividing each side of this equation by **152** yields **18** $= h$. Therefore, the height, in inches, of the prism is **18**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.